Order of Operations with Radicals

Lesson Plan

1. Lesson Information

Course	Grade 7 Accelerated (1205050)	Unit	4 — Solving Problems with Rational Numbers
Lesson	Lesson 6 — Order of Operations with Radicals	Date	_____
Length	53 minutes (one period)	Benchmark(s)	MA.8.NSO.1.7

2. Standards

MA.8.NSO.1.7 Solve multi-step mathematical and real-world problems involving the order of operations with rational numbers including exponents and radicals.

Must. Evaluate multi-step expressions with rational numbers, exponents and radicals, in six steps or fewer.

Must not. No mnemonic. The state guide says to avoid PEMDAS because it leaves out grouping symbols, and the word appears on no student page in this unit.

3. Mathematical Thinking and Reasoning Standards

the evidence for each, from this period

MTR	What it asks	The evidence in this lesson
MA.K12.MTR.5.1	Use patterns and structure to help understand and connect mathematical concepts.	Notes I — the four levels are read off the expression's structure and the class is never given a mnemonic, because the guide asks us not to give one.
MA.K12.MTR.3.1	Complete tasks with mathematical fluency.	Boards 1, 3 and 7 — one structural decision each, written and up on the cue; these are the errors that cost marks everywhere else.
MA.K12.MTR.4.1	Engage in discussions that reflect on the mathematical thinking of self and others.	Board 5 — when the room splits between 50 and 80, the sixty-second re-vote before the reveal.

4. Learning Target and Essential Question

Learning target. I can evaluate a multi-step expression with rational numbers, exponents and radicals by reading its structure.

Essential question. What decides the order in which an expression is evaluated?

Building on. 4.05 — the radical bar as a grouping symbol; 3.04–3.05 — negative exponents on integer and rational bases; Grade 6 — the order of operations with whole numbers.

Working toward. 4.07–08 — the same structure inside real-world problems; MA.912.NSO.1.4 in Algebra 1.

5. Sequence

read from the deck; never typed

Segment	Min	What the teacher does	What students do
Settle in; post the learning target	1	Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet.	Copy the target.
Warm-Up	6	Two minutes silent, then reveal. One sentence of reteach per question at most. Then the new words, read once.	Four retrieval questions, alone, no notes. Then copy the new words.
Notes	12	Teach from the front. Expand once per law; say the OFF line out loud.	Copy the notes; answer the checks aloud.
Example 1	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Example 2	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Whiteboards	11	One question at a time. Boards down until the cue, all up together; scan the back row first.	Answer on a board; hold it up on three.
Independent Set	6	Silent. Circulate and mark what you see; do not teach.	Six questions, written, alone.
IXL	5	Assign the day's IXL skills.	Start the skills; finish tonight.
Total	**53**		

6. Gradual Release

I do. The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the OFF line is said out loud rather than printed.

We do. Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

You do. The nine-question whiteboard round (11 minutes) and then the six-question independent set, silent and written.

7. Higher-Order Questions

Question	DOK
Why is -2^2 different from $(-2)^2$? What is the exponent attached to in each one?	DOK 3
A mnemonic for the order of operations has no letter for the radical bar or the fraction bar. What does reading the structure give you that the mnemonic cannot?	DOK 3
Roscoe says a square is always positive. What would have to be true about $\sqrt{n} - 11$ for that to be right, and what value of n breaks it?	DOK 3

8. Checks for Understanding

each board, and what to do about it

#	What the board asks	Answer	If the board is wrong
1	$90 \div 5 \cdot 3$	***54***	6 — $5 \cdot 3$ was done first [8.NSO.1.7: multiplication and division are one level, taken left to right]; 1,350 — the division was read as a multiplication
2	A square tile has side √196 centimeters. Its diagonal is about 19.8 centimeters. What is the perimeter of the tile?	***56 centimeters***	anything built from 19.8 — the unneeded figure was used [a perimeter needs the side]; 14 — the side, not the perimeter; 196 — the root was never taken
3	$16 - 5 + 4$	***15***	7 — $5 + 4$ was done first [8.NSO.1.7: addition and subtraction are one level, taken left to right]
4	$\frac{3^3 - 21}{5^2 - 15}$	***3/5***	anything that divides before the top and bottom are finished [8.NSO.1.7: the fraction bar groups both]; −6/5 — 5^2 read as $5 \cdot 2$, leaving 6 over −5; 6/25 — the −15 under the bar was dropped
5	$2 + 3 \cdot 4^2$. Which is the value?	***B — 50***	(A) the exponent was applied to $3 \cdot 4$ instead of to the 4 alone [8.NSO.1.7: an exponent belongs to its own base]; (C) 4^2 was read as $4 \cdot 2$ [8.NSO.1.3: an exponent is not a factor]; (D) $2 + 3$ was done first, then $5 \cdot 16$ [8.NSO.1.7: multiplication comes before addition]
6	$(6^2 - 4)$ over 2^3. Which is the value?	***A — 4***	(B) only the 4 was divided by 2^3 before the subtraction [8.NSO.1.7: the bar groups the whole numerator]; (C) 6^2 was divided by 2^3 first and then 4 was subtracted [8.NSO.1.7: the bar groups the whole numerator]; (D) the numerator was multiplied by 2^3 instead of divided [8.NSO.1.7: the bar is division]
7	$-2^2 + (-2)^2$	***0***	8 — both terms read as $(-2)^2$ [8.NSO.1.7: an exponent attaches to its own base, and parentheses decide what that base is]; −8 — both terms read as -2^2
8	$12\left(\left(\frac{1}{2}\right)^3 + 3^{-2}\right)$	***17/6***	12/8 + 12/9 left unadded — a value, not an answer; 1/8 + 1/9 = 2/17 — the denominators were added [Grade 5: adding fractions needs a common denominator]
9	Roscoe says that $(\sqrt{n} - 11)^2$ is always positive, because a square is always positive. Is he right? Give a value of n that settles it.	***No — at n = 121 the value is 0***	'yes, squares are positive' — the claim repeated rather than tested [8.NSO.1.7: a square is never negative, which is a different statement]; 'no' with no value of n — the counterexample is the evidence

MTR.4.1 — the one peer move. When a board splits — no option holding about two-thirds of the room — sixty seconds of “convince the person next to you”, then re-vote, then reveal. The minute comes out of the whiteboard remainder.

9. Differentiation

ESE / IEP. Give the four-level table from Notes I as a card and let it stay out. On every expression the student first circles all the grouping symbols and numbers them from the inside out, and only then evaluates; the circling is the work, and the arithmetic follows it.

ELL. Grouping symbol is the phrase to own. Put the four of them on the board — parentheses, brackets, the radical bar, the fraction bar — and have the student point to each one in an expression and name it before evaluating anything.

Enrichment. Ask for an expression using 2, 3 and 4 once each, with any grouping symbols, whose value is as large as possible — then as small as possible. Then ask why a mnemonic could not have answered either question.

10. Closure, Vocabulary, Materials, Practice

Closure. Board 9 is written work and it is the exit evidence: the student has to give a value of n that settles Roscoe's claim and say what is wrong with the sentence, not only that it is wrong. Read the boards, not the papers.

Vocabulary. **grouping symbol**; **nested**; **multi-step expression**

Materials. Whiteboards and markers. Calculators are allowed on everything, and this is the lesson where a calculator is worth using against the class: enter -2^2 and $(-2)^2$ and let the machine settle it.

Practice. The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due at the start of the next class: Evaluate radical expressions — D8J.